

Data-Driven Optimization of Sugarcane Milling Processes

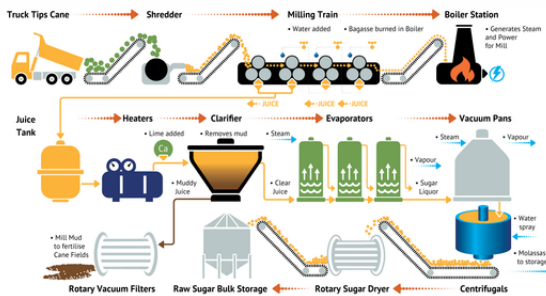


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GQP Sponsors: Sagar Mahurkar, Prachi Mahurkar



Motivation & Background

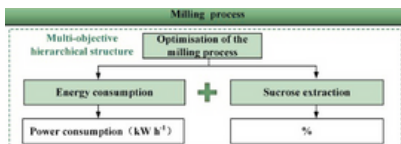
Sugarcane is the world's largest agricultural crop with a global production of ~2.0 billion tons annually.



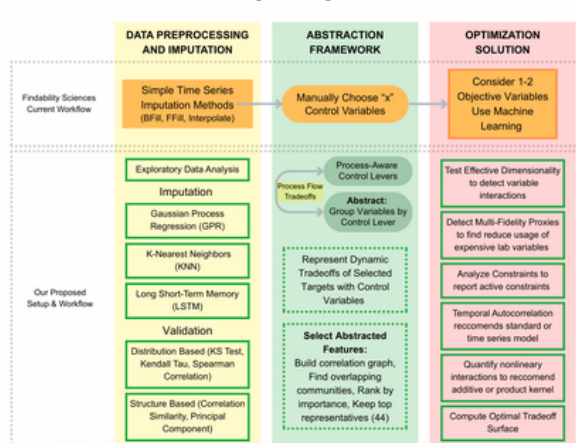
- Supports ethanol biofuel production
- Contributes to renewable energy sources
- Bagasse reused as fuel in mills

Problem Statement

How can multi-objective optimization be used to balance energy consumption and sugar yield in the milling process?



Workflow

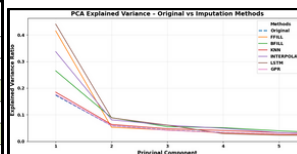


Methodology + Results

Imputation

Imputation Methods were validated off
Gaussian Process Regression (GPR) used a RBF + White Kernel
K-Nearest Neighbors with n=5 and used nan-euclidian distance

Imputation Method	KS Similarity (%)	Correlation RMSE	Correlation Similarity	PCA Similarity	Spearman Similarity	Kendall Similarity
BFILL	87.68	0.189	0.772	0.9919	0.7398	0.7421
FFILL	88.01	0.2974	0.7434	0.9462	0.7293	0.7417
GPR	77.47	0.0939	0.9396	0.9999	0.8713	0.8768
INTERPOLATE	87.06	0.2272	0.7698	0.9784	0.7412	0.7469
KNN	95.57	0.0818	0.9465	0.9992	0.935	0.9337
LSTM	82.88	0.3472	0.6464	0.9523	0.6431	0.6513



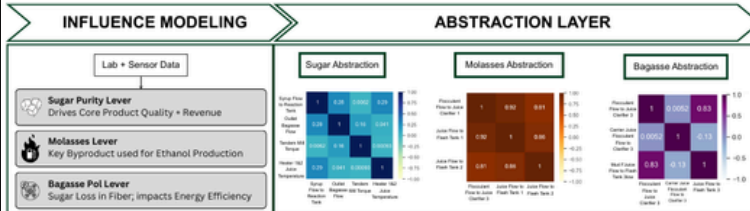
Abstraction

Influence Modeling:

Identify primary control levers
Map strongest process-variable relationships

Abstraction Layer:

Process-Level Abstractions Derived From Control Levers

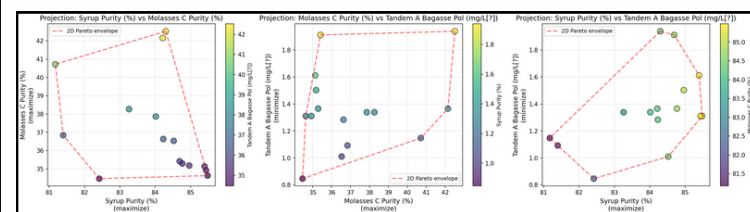


Optimization

Testing:

- Effective dimensionality - determines if relationships are additive
- Multi-fidelity proxies - finds cheap sensor proxies for expensive lab tests
- Constraint analysis - identifies active constraints
- Temporal autocorrelation - checks time dependencies
- Interaction strength - quantifies variable interactions

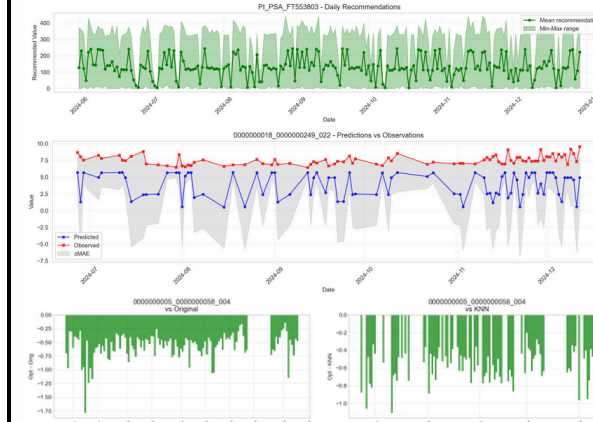
Full GP Model, found one proxy, 41/49 controllable variable constraints, Time-series model, and additive kernel between variables.



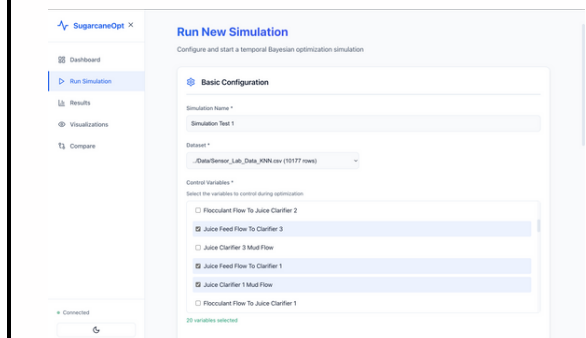
Product

Simulation Results

Our simulation had Predictive accuracy for the best recommendations - MAE: 0.5281, SAE of 0.4439
Averaged Improvement of 4.837% throughout 24-25 season



Application Prototype



Recommendations & Future Work

- Apply advanced feature clustering & causal modeling
- Adopt factor analysis for abstraction
- Deploy Autonomous Optimization on Milling Process

Acknowledgements

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